# Validation-first EEG pipelines: how synthetic benchmarks mislead, and how pre-registration catches it

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October 2026

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## Abstract

Automated EEG analysis pipelines are usually developed and evaluated on the same synthetic or public data, and the outcome measures are chosen after the results are known. We built an open, deterministic scalp-EEG pipeline (quality control, zero-phase filtering, aperiodic/periodic spectral parameterisation, debiased weighted phase-lag index, an unsupervised rhythmic-event detector and a citation-checked plain-language summary) under a validation-first protocol: six hypotheses with numeric pass thresholds were committed to version control before the code or the data they concern existed, every method was tested against synthetic ground truth, and the test suite and the explanation checker were themselves mutation-tested. Four of the six hypotheses were rejected. The protocol's value lay in how they failed. A synthetic detection benchmark on which the detector passed (sensitivity 0.955 at 0.69 false detections per hour) hid four failure modes that only appeared on real recordings or on deliberately harder synthetic negatives: a rhythm gate fooled by the 1/f background (the very confound one hypothesis was written to test), harmonics of spike-wave discharges, normal alpha and spindle bursts, and blink trains merged into "rhythms". On a harder benchmark the detector fell to 0.847 [95 % CI 0.803-0.882] at 1.68 false detections per hour, and on a pre-registered held-out subject of the CHB-MIT database it found 4 of 7 seizures at 3.3 false detections per hour. The pre-registered estimator thresholds also exposed the small-sample bias of plain wPLI (95th percentile 0.19 under pure volume conduction at two minutes), and a mechanical claim checker rejected all 2157 live mutated explanations once a rounding-tie defect found on its first run was fixed. We argue that, for methods papers, the benchmark itself is a researcher degree of freedom, and we give a short, checkable protocol to constrain it. Code, data-retrieval scripts with checksums and all result files: https://doi.org/10.5281/zenodo.23447346.

**Keywords:** EEG, validation, pre-registration, synthetic data, aperiodic activity, phase-lag index, seizure detection, reproducibility, mutation testing

## 1. Introduction

The same EEG recording analysed by different teams gives different answers (Trübutschek et al. 2024), and preprocessing choices alone change downstream results (Robbins et al. 2020). Methods papers usually respond with a new pipeline and a benchmark on which it performs well. The benchmark is rarely fixed before the method is tuned, and when it is synthetic, it contains exactly the phenomena its author thought of. Leakage between development and evaluation data is a known source of irreproducible

results in machine-learning-based science (Kapoor and Narayanan 2023), and pre-registration is the standard remedy in empirical research (Nosek et al. 2018). Neither practice is routine in signal-processing methods work, where the “experiment” is a simulation the author designs.

This paper reports what happened when an EEG pipeline was built under a validation-first protocol, and treats the pipeline mainly as the case study. Three well-documented interpretation errors motivated it: a change in canonical band power read as a change in an oscillation when only the aperiodic (1/f) exponent moved (Donoghue et al. 2020); volume conduction read as connectivity (Nolte et al. 2004; Vinck et al. 2011); and, newer, plain-language summaries that state numbers nobody computed. For each, a hypothesis with a numeric threshold was written down first, and for the event detector, two more tested whether synthetic validation transfers to real data.

Our contributions are:

1. an account of how a synthetic benchmark misled the development of an event detector four times, and which protocol element exposed each failure (Section 3.2);
2. evidence that pre-registering the *estimator*, not only the measure, matters, from the small-sample bias of wPLI (Section 3.3);
3. a citation contract for generated explanations, and a mutation-testing evaluation of the checker that enforces it (Section 3.5);
4. a short protocol for validation-first methods work that a reviewer can check from a repository (Section 4.1).

## 2. Methods

### 2.1 Pipeline

The pipeline reads EDF/EDF+, BDF or BrainVision files through MNE-Python, refuses oversize or unsupported files before parsing, converts to microvolts and drops identifying header fields. Flat channels are marked on the raw data; the signal is then filtered with a Hamming-window FIR design applied in a single pass with delay compensation (zero phase, magnitude response $H$ rather than the $H^2$ of forward-backward filtering; Widmann et al. 2015): a 50 Hz notch with harmonics and a 0.5-45 Hz band-pass. Noisy and low-amplitude channels are marked with median/MAD statistics, and referential montages are re-referenced to the common average of good channels (bipolar montages are left unchanged).

Events are detected before spectra are estimated, so that spectra and coupling describe the background. The detector computes line length (Esteller et al. 2001) on the 1-20 Hz band in 2 s windows (1 s step), converts it to a robust z-score per channel (median/MAD over the recording), requires a fraction of channels to exceed a threshold, keeps runs of at least a minimum duration, merges runs separated by less than a gap, and finally gates each candidate on the fundamental frequency of its spectrum divided by the background spectrum of the same channels. All five thresholds were chosen by grid search on synthetic tuning seeds only (Section 2.3).

Spectra are Welch estimates (2 s Hann windows, 50 % overlap) excluding event windows. The aperiodic and periodic components are separated with a specparam-style model

(Donoghue et al. 2020): a robust line in log-log space, greedy Gaussian peak detection and a joint bounded least-squares fit; specparam itself is used only as a test oracle. Fits with $R^2$ below 0.8 or an exponent outside [0, 4] are listed and excluded from averages. Phase coupling is reported as the debiased squared weighted phase-lag index (dwPLI; Vinck et al. 2011), with the phase-locking value (PLV) shown only as a contrast.

The output is a JSON file of metrics that is byte-identical across runs, a set of figures, and an HTML report with a plain-language explanation (Section 2.5).

### 2.2 Pre-registration

Hypotheses H1-H5 were committed to the repository before the code that tests them (commit `1d92eb4`, 26 minutes before the first code commit). H6 was committed after H5 failed and before any file of its test subject was downloaded (`ed41812`); the detector was frozen in a later commit (`36b7964`) and run once on that subject. Each hypothesis states the data, the measure and the numeric threshold:

| | hypothesis | pass condition |
|---|---|---|
| H1 | aperiodic confound | exponent +0.5, alpha unchanged: 8-12 Hz band power changes ≥ 30 % in ≥ 95 % of cases, and fitted alpha peak power changes ≤ 0.05 log10 in ≥ 90 % |
| H2 | synthetic detection | 3 Hz spike-wave at ≥ 0 dB SNR: sensitivity ≥ 0.90, Wilson lower bound ≥ 0.80, ≤ 2 false detections/h |
| H3 | volume conduction | independent sources mixed into 19 sensors: PLV > 0.3 in ≥ 50 % of pairs; wPLI 95th percentile ≤ 0.15; a truly lagged pair has wPLI > 0.5 |
| H4 | explanation faithfulness | checker rejects every live mutant (≥ 300 from ≥ 6 operators) and accepts every faithful explanation |
| H5 | transfer (exploratory) | CHB-MIT chb01: ≥ 5 of 7 seizures at ≤ 2 false/h, thresholds from synthetic data only |
| H6 | held-out subject | CHB-MIT chb03, frozen detector, one run: ≥ 60 % of seizures at ≤ 2 false/h |

A rejected hypothesis is reported as rejected; changes made after a result are described as exploratory and, where they affected a later test, that test was re-registered on new data. One deviation is recorded: the tuning rule's false-detection constraint was relaxed from 1 to 2 per hour, H2's own limit, after no grid point met it; the infeasible grid is kept as a result file.

### 2.3 Synthetic ground truth

Recordings are generated with a 19-channel 10-20 layout at 256 Hz. Each channel's background is shaped noise with a random aperiodic exponent and offset and an optional alpha peak; flat and noisy channels, 50 Hz mains interference, blinks (frontal, about 15 per minute), muscle bursts, posterior alpha bursts and spindles can be added. Events are 3 Hz spike-wave trains with a small frequency glide, injected into a random subset of channels at a set SNR (0, 3, 6 or 10 dB). For H3, independent sources are mixed into sensors through a random lead field, with and without one truly lagged source pair. The generator writes the ground truth to a separate file that the pipeline never reads.

Tuning seeds (0-59) and evaluation seeds (1000-1199) are disjoint. In the final benchmark one recording in five contains no events. The benchmark evolved during development (Section 3.2); every version is in the repository history.

### 2.4 Real data

The CHB-MIT Scalp EEG Database v1.0.0 (Shoeb 2009; Goldberger et al. 2000), paediatric recordings in a bipolar montage at 256 Hz, was retrieved from PhysioNet's open-data mirror with SHA-256 verification. A pre-registered file rule selected, for each subject, every file with at least one seizure plus the first three seizure-free files. Subject chb01 (7 seizures, 9.6 h) was the H5 test and then became a development set; subject chb03 (7 seizures, 10 h) was the held-out H6 test. Detections were scored with the any-overlap rule (Ziyabari et al. 2017); false detections per hour are reported with a Poisson upper bound.

### 2.5 Explanations and the claim checker

The plain-language summary is written either by a deterministic template or, optionally, by a large language model that receives only the metrics. In both cases every number must carry a citation to the metric it comes from, written `value [[metric.key]]`. A checker that contains no model parses the text with regular expressions and rejects it if a cited key does not exist, a written value does not round to the cited metric at its written precision (with a strict rule at exact ties), a number has no citation, a channel name is not in the recording, diagnostic vocabulary appears, or the research-use notice is missing. Rejected text is replaced by the template and the failure is shown in the report.

The checker was evaluated by mutation: nine operators (perturb a number, scale it by 1000, swap a citation for another key, cite a non-existent key, add an uncited number, invent a channel, add a diagnosis, drop the notice, and drop a citation) applied four times each to 60 faithful explanations of varied synthetic recordings. A mutant whose text is still true at the written precision is "equivalent" and excluded; equivalence is decided from the metric values by integer rounding, independently of the checker. The same principle, validating the validator with realistic wrong answers and mutation testing, was applied earlier to automated graders for AI agents on biomedical engineering tasks (Mbote 2026).

### 2.6 Testing the tests

Beyond unit, property-based (linearity, time-shift equivariance), metamorphic (Parseval), differential (specparam) and ground-truth tests, seventeen one-line code mutants (for example, a causal filter, a forward-backward second pass, plain instead of debiased $wPLI^2$, a nondeterministic value in the output) were run against the test suite; a surviving mutant marks a blind spot.

### 2.7 Statistics and theory checks

Proportions carry Wilson 95 % intervals, zero-failure rates the rule of three, and false-detection rates a Wilson-Hilferty Poisson upper bound. Where a decision rested on a statistical model, the model's prediction was compared with the measurement: the Welch sampling distribution for H1, the null distribution of the imaginary cross-spectrum for wPLI bias, Whittle maximum likelihood for the log-space fit, the Fourier transform of a blink train for the gate floor, a Poisson process for blink merging, a contamination model for robust z-scores, and binomial power for H6.

All computation ran on one CPU core of a laptop with 8 GB of RAM; a one-hour, 32-channel recording is processed in 12.7 s with 1.5 GB peak memory.

## 3. Results

### 3.1 Confirmatory outcomes

| | result | verdict |
|---|---|---|
| H1 | band power changed ≥ 30 % in 200/200 cases (median −68 %); peak power within 0.05 log10 in 60.5 % | rejected |
| H2 | sensitivity 0.847 [0.803, 0.882] (271/320 events), 1.68 false/h (56 in 33.3 h) | rejected |
| H3 | PLV > 0.3 in 92 % of pairs; wPLI p95 0.193 at 2 min, 0.095 at 10 min; lagged pair 0.998 | rejected at 2 min, accepted at 10 min |
| H4 | 2157/2157 live mutants rejected, 60/60 faithful accepted | accepted (rejected on the first run) |
| H5 | 2/7 seizures, 0.52 false/h | rejected |
| H6 | 4/7 seizures (Wilson [0.25, 0.84]), 3.30 false/h (upper bound 4.6) | rejected |

Engineering requirements were met: aperiodic exponent median absolute error 0.014 on 200 synthetic spectra, agreement with specparam in 200/200, alpha peak frequency error 0.067 Hz, and perfect recall and precision for bad channels on 100 recordings.

### 3.2 How the synthetic benchmark misled

The detector's development is a sequence of four failures, each invisible to the benchmark of the moment.

**The 1/f trap, rebuilt inside the detector.** The first detector gated candidates on the dominant frequency of their raw spectrum. Synthetic discharges were strong enough to dominate it; it passed the synthetic benchmark. On chb01 all seven seizures passed the line-length stage, but the raw spectra of real EEG peak at 1-2 Hz because of the aperiodic background, so the gate rejected five (H5 rejected, 2/7). This is the confound H1 was written to test, reproduced in our own code. Dividing the event spectrum by the background spectrum of the same channels recovered the rhythm (2.5-3.5 Hz in five of seven seizures). Because chb01 had now informed a design change, it became a development set and a new held-out test (H6) was registered on a subject not yet downloaded.

**Normal rhythms.** With the normalised gate, chb01 gave 5/7 seizures but 51 false detections (5.3/h), most at 8-13 Hz: posterior alpha bursts and spindles, which rise above a recording's own background exactly as seizures do. The benchmark contained no such bursts. They were added to the generator first, so that the synthetic tuning rule, not chb01, chose the response: a delta-theta gate.

**Harmonics.** With normal rhythms in the benchmark, no grid point met the tuning constraint. Listing every candidate showed that dividing by a 1/f background lifts a spike-wave train's harmonics above its fundamental; the gate now reads the lowest peak at least half the size of the largest.

**Blink trains.** On event-free recordings the detector made 6.4-6.6 false detections per hour at every threshold. A blink band-limited to 1-20 Hz is a wave of about 2 Hz, and at 15 blinks per minute a 5 s merge gap joined consecutive blinks into long “rhythmic” events. The benchmark hid this because every tuning recording contained real events, which raised the robust baseline. A Poisson model of supra-threshold blinks predicts 13.7 long merged runs per hour at a 5 s gap and 0.07 at 1 s (measured 21.9 and 0.3). One recording in five is now event-free and the gap is tuned.

The cost of realism was measured on the same detector family: the first campaign accepted H2 on the earlier benchmark (sensitivity 0.955 at 0.69/h; result file at commit `19198f7`), and the final campaign rejected it (0.847 at 1.68/h; 0.95-0.96 at 6-10 dB, 0.71-0.76 at 0-3 dB). We report the final run as the verdict because the earlier benchmark lacked negatives known to occur in real EEG; both runs are in the repository. On held-out chb03 the frozen detector found 4 of 7 seizures at 3.30/h (H6 rejected). The three seizure-free files produced no false detection; every false detection fell in a file containing a seizure. All three missed seizures exceeded the line-length threshold (peak z 5.8-12.2) but had fundamentals of 1.0-1.5 Hz, below the gate’s 2 Hz floor, a floor that exists to reject blinks (post hoc diagnosis; no change made). On chb01, now a development set, the same detector finds 6/7 at 1.66/h, a number that is optimistic by construction.

### 3.3 Pre-register the estimator, not only the measure

Under pure volume conduction (six independent sources, no coupling), PLV exceeded 0.3 in 92 % of sensor pairs, as expected. Plain wPLI, the measure named in H3, had a 95th percentile of 0.193 with two minutes of data (60 epochs), above the registered 0.15, and 0.095 with ten minutes. The bias is predicted by the null distribution of the imaginary cross-spectrum: expected wPLI 0.1457 at 60 epochs (measured 0.1455) and 0.0651 at 300 (measured 0.0655). The debiased estimator stayed at 0.021 and 0.006 and still detected a truly lagged pair (median 0.998). H3 is rejected at two minutes as registered; the pipeline reports dwPLI.

### 3.4 The aperiodic confound holds; the registered tolerance did not

When only the aperiodic exponent rose by 0.5, 8-12 Hz band power fell by a median of 68 % and changed by at least 30 % in all 200 cases, with the oscillation unchanged; rotating about 20 Hz instead of 1 Hz raised band power by a median of 44 %, so the sign of a band-power change is not informative either. Fitted peak power, however, stayed within 0.05 log10 in only 60.5 % of cases. A paired analysis (same noise realisation in both conditions) puts the method’s bias at 0.0012 log10. The Welch sampling distribution predicts a per-bin log-power noise of 0.0334 (measured 0.033) and a peak-height difference SD of 0.038 for an efficient estimator; the measured SD was 0.057, so the estimator runs about 1.5 times above the sampling floor. H1’s tolerance was tighter than two separate three-minute recordings allow. It was a design error in the hypothesis, recorded as a rejection rather than corrected after the fact.

### 3.5 The claim checker

On its first full run the checker caught 2152 of 2153 live mutants. The escape wrote “1” while citing a metric equal to 0.5: the checker accepted the tie, and the independent equivalence oracle (round half to even) did not. Both now use a strict rule, and the

template adds a decimal rather than write a tie. After the fix, 2152/2152 were rejected, and after a later revision of the template wording 2157/2157 (miss-rate upper bound 0.0014), with 60/60 faithful explanations accepted. Two limits remain. The checker verifies that each number traces to a computed value, not that the cited key fits the sentence: a channel count citing a filter cutoff of the same value passes. And numbers written as words are outside the contract (60/60 such mutants passed). These are stated in every report.

### 3.6 The test suite's blind spots

Three of seventeen code mutants survived the first run: a forward-backward second filter pass (no test measured the output at the cutoff), plain instead of debiased wPLI$^2$ (both are small under the null), and a timestamp in the output that rounding made identical across quick runs. Each led to a new test; the final run killed 17 of 17.

## 4. Discussion

The detector is not the contribution. A per-recording, unsupervised line-length detector with fixed thresholds is weak; patient-specific detectors on CHB-MIT report far higher sensitivity (Shoeb 2009). What the protocol produced is an evidence trail in which every failure was found, dated, explained and kept.

Three observations generalise. First, a synthetic benchmark is a hypothesis about what matters, and it encodes the developer's blind spots: each of the four detector failures was absent from the benchmark until a real recording or a deliberately adversarial negative revealed it, and the headline sensitivity fell from 0.955 to 0.847 as the benchmark became more realistic. The benchmark is therefore a researcher degree of freedom in the same sense as an analysis choice, and its versions should be reported. Second, pre-registration did not prevent mistakes; it made them unambiguous. Without a committed threshold, the wPLI bias and the H1 tolerance would have read as "acceptable" results, and the chb01 numbers after redesign (6/7) would have been reported as a test. Third, the cheapest controls caught the most: event-free recordings, a held-out subject named before download, and mutants of the tests themselves.

### 4.1 A checkable protocol

We suggest the following minimum for methods papers whose evaluation is partly synthetic. Each item can be verified from a repository.

1. Hypotheses with numeric thresholds, naming the estimator, committed before the code and data they concern.
2. Disjoint tuning and evaluation seeds; the benchmark's generator and every version of it in version control.
3. Negatives the method is likely to confuse with its target, and records with no target at all.
4. A real-data subject held out and named before download; any subject that informed a design change relabelled as development.
5. Theory-based predictions for the key estimators, compared with simulation.
6. Mutation testing of the test suite and of any automated checker.

7. Rejected hypotheses reported as rejected; deviations listed.

### 4.2 Limitations

Pre-registration here relied on commit history in the same repository, which a determined author could rewrite; an external registry such as OSF would be stronger and is planned for the next round. The work was done by one person over a short period, so decisions were not independently reviewed. The real-data base is small (two paediatric subjects, about 20 hours, 14 seizures, one used for development). With seven seizures, H6 was close to a coin flip by design: a detector with true sensitivity 0.60 passes its rule with probability 0.42. Showing a lower bound above 0.6 with 80 % power would take about 76 seizures if the true sensitivity were 0.75. The synthetic generator lacks sleep stages, interictal spikes and evolving ictal frequency content. The language-model explainer was implemented but not evaluated against a live model in this version, so H4 concerns the checker, not a model's behaviour. The tool is for research use only and is not a medical device.

### 4.3 Future work

A spatial cue to separate frontal, symmetric blinks from slow ictal delta, which would allow a lower gate floor; a confirmatory test sized by the power analysis on further CHB-MIT subjects; and an evaluation of language-model explanations, measuring first-pass acceptance and repair rates under the checker.

## 5. Conclusion

Building an EEG pipeline against pre-registered thresholds rejected four of six hypotheses and exposed four failures that a synthetic benchmark had hidden, including the exact confound the project was designed to avoid. None of these would have been visible in a conventional benchmark-and-report workflow. Validation-first development costs little computation and makes methods results harder to overstate.

## Data and code availability

Source code, the synthetic generators, the data-retrieval script with checksums, every result file with provenance and a SHA-256 manifest, and the build journal are archived at https://doi.org/10.5281/zenodo.23447346 (release v0.1.1, MIT licence) and developed at https://github.com/mbote-droid/Neuro-signal. CHB-MIT is available from PhysioNet under the Open Data Commons Attribution License v1.0.

## Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

## Competing interests

The author declares no competing interests.

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